

Appropriability and the Market Pricing of AI Model Capability*

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Abstract

Do financial markets price the capability of large language models (LLMs), or do they merely react to AI hype? We study 60 major LLM release events from 2024 to early 2026, constructing a firm-event panel of 86 AI-related listed companies. We measure model capability using the Artificial Analysis Intelligence Index and estimate cumulative abnormal returns over a 20-trading-day window. Closed-source model releases drive a positive capability-pricing effect: a one-standard-deviation increase in intelligence is associated with a 3.0 percentage point higher $CAR[0, +20]$ ($\hat{\beta} = 0.0023$, $p = 0.008$ under wild cluster bootstrap). This effect is absent for open-source releases, where the capability slope is attenuated and reverses sign in point estimates. Media sentiment, measured by FinBERT, is negatively associated with post-release returns—consistent with narrative-driven pre-pricing that is distinct from technological capability. Markets appear to price AI capability primarily when it can be appropriated.

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1 Introduction

Large language model releases have become routine technology shocks—roughly 30 major releases per year in 2024–2025—attracting widespread investor attention. Less clear is whether markets price the *capability* embedded in these releases or merely the *narrative* surrounding them. This distinction matters for understanding how financial markets process technological information and for valuing firms whose competitive positions depend on appropriable AI performance.

Appropriability theory predicts that the market value of a technological advance should depend on whether the innovating firm can capture the rents it generates (?). Closed-source LLM deployments create proprietary barriers—API lock-in, ecosystem advantages, and pricing power over model access. Open-source releases, by contrast, diffuse capability broadly and may commoditize inference services, eroding the competitive rents that would otherwise be capitalized into stock prices. If appropriability determines market pricing, the coefficient on model capability should be positive for closed-source releases and weaker—or reversed—for open-source releases.

We test this prediction using a firm-event panel spanning 60 major LLM releases from January 2024 through early 2026. We measure capability with the Artificial Analysis (AA) Intelligence Index, a composite benchmark score that aggregates performance across standardized reasoning and knowledge tasks. Our main outcome is the market-model cumulative abnormal return over $[0, +20]$ trading days, estimated separately for each firm-event pair and then regressed on the AA Intelligence Index, conditioning on firm-level controls (size, book-to-market, volatility, momentum) and year fixed effects. We cluster standard errors by event and report results under both CR2 (Bell-McCaffrey small-cluster correction) and a manually implemented Rademacher wild cluster bootstrap.

Three findings emerge. First, among closed-source releases, capability is positively priced with strong statistical support: $\hat{\beta} = 0.0023$, CR2 $p = 0.007$, wild bootstrap $p = 0.008$, over 36 event clusters. A one-standard-deviation increase in the Intelligence Index is associated with

a 3.0 percentage-point higher 20-day abnormal return. Second, when we interact capability with an open-source indicator, the slope for open-source releases is substantially attenuated in point estimates (implied slope = -0.0015), though the interaction term is only marginally significant under the most conservative inference (CR2 $p = 0.048$, wild bootstrap $p = 0.086$); results should be read as suggestive. Third, media sentiment measured by FinBERT is *negatively* associated with $\text{CAR}[0, +20]$ ($\hat{\beta} = -0.081$, CR2 $p = 0.003$), and intelligence remains positively predictive after controlling for sentiment, suggesting the capability effect reflects technological information rather than narrative heat.

This paper connects to three bodies of work. Event studies on technology shocks document positive abnormal returns around IT capability announcements (, CITATION)), but they cannot separate capability from attention because both co-move with the event. ? show that patent citations—a proxy for technological appropriability—predict firm market value, establishing that rent-capture moderates the value of innovation. ? and ? document that media sentiment moves short-run prices; however, they do not test whether fundamental capability survives sentiment controls or examine the direction of long-run returns. No study we are aware of isolates *which dimension* of AI model performance drives post-release stock returns or whether the capability-to-value link depends on the appropriability of the release. We provide this evidence, using direct capability scores rather than proxies, and decompose LLM event returns into a fundamental-capability component and a narrative component that move in opposite directions.

The remainder of this paper proceeds as follows. Section ?? describes the data and variable construction. Section ?? develops two testable hypotheses. Section ?? presents the main results. Section ?? reports robustness checks. Section ?? concludes.

2 Data and Variable Construction

Event sample. We compile 60 major LLM release events from January 2024 through March 2026, drawing on model release announcements documented in AI benchmarking databases and verified against official developer communications. Events are aggregated to the model-family level when a developer releases multiple variants simultaneously; a release event is included only if it corresponds to a meaningfully new capability level rather than a minor update. The result is one observation per release, with a confirmed public announcement date.

Firm-event panel. For each event we include up to 86 AI-related listed firms whose business activities are plausibly connected to LLM development or deployment. Connection types include direct model development (owner), equity investment in the developer (investor), cloud infrastructure provision (cloud), upstream hardware supply, downstream software integration, and direct competition. We identify these relationships through public filings, investment records, and product documentation, and we code them as binary pre-event indicators. We control for firm-model exposure using these pre-event relationship indicators; full variable definitions are in Appendix A. The full panel has 5,161 firm-event observations; the regression sample with non-missing Intelligence Index and controls contains 3,780 observations across 47 event clusters.

Cumulative abnormal returns. We estimate daily abnormal returns using a market model with parameters estimated over a $[-200, -21]$ pre-event window (?). We then accumulate abnormal returns over post-announcement windows, with the primary window $[0, +20]$ (twenty trading days). The event date is the verified public announcement date.

Model capability. The AA Intelligence Index is a composite score produced by Artificial Analysis that aggregates performance across mathematics, reasoning, and general knowledge benchmarks. It has a mean of 26.6 and a standard deviation of 13.1 in our regression sample.

We center the index at its sample mean (*intel_c*) so that main effects of binary moderators are interpretable at average capability levels.

Open-source indicator. We code *is_open_weight* = 1 if the model is released under an open-weight or fully open-source license permitting free redistribution and inference, and 0 otherwise. Among the 60 events, 11 are open-source (881 firm-event observations) and 36 have non-missing intelligence data under a closed-source designation (2,902 observations).

Media sentiment. For each event we compute the mean FinBERT-based sentiment score of news articles published in a ± 5 -day window around the announcement date. The sentiment variable has a mean of 0.18 and a standard deviation of 0.16, indicating that AI model releases attract generally positive coverage. We center sentiment at its sample mean (*sent_c5*).

Controls. Firm-level controls are the log of total assets, the book-to-market ratio, realized return volatility over the prior quarter, and momentum (prior 12-minus-1-month return). Year fixed effects absorb aggregate time trends. All controls are measured as of the quarter preceding the event.

3 Hypothesis Development

Appropriability theory (?) holds that a technological advance creates stock market value only to the extent that the innovating firm—or firms positioned to benefit—can appropriate the rents it generates. Applied to LLM releases, this logic yields two testable predictions.

H1 (Closed-source capability pricing). *Model capability is positively associated with post-release cumulative abnormal returns when the model is closed-source.* A higher-capability closed-source model raises the prospect of expanded API revenue, stronger customer lock-in, and a wider competitive moat.

H2 (Open-source attenuation). *The capability-pricing slope is weaker for open-source releases*

than for closed-source releases. Open-weight models allow unrestricted use and redistribution, eroding proprietary barriers. Note that the prediction concerns *relative slopes*, not the unconditional average return.

We additionally examine whether media sentiment explains the capability result. If intelligence merely reflects AI hype, controlling for sentiment should absorb the intelligence coefficient. The alternative—that intelligence reflects technological information beyond sentiment—implies that both retain independent predictive power.

4 Results

4.1 Baseline capability pricing

Table ?? presents the baseline results. Among closed-source releases, capability is strongly and precisely priced: $\hat{\beta} = 0.0023$ (SE = 0.0006), CR2 $p = 0.007$, wild bootstrap $p = 0.008$, over 36 event clusters. **A one-standard-deviation increase in the AA Intelligence Index (13.1 points) is associated with a 3.0-percentage-point higher 20-day abnormal return for firms exposed to a closed-source release.** The result survives year fixed effects and the full set of firm-level controls, consistent with H1.

The full-sample estimate is positive but weaker: $\hat{\beta} = 0.0015$ (SE = 0.0007), CR0 $p = 0.027$, CR2 $p = 0.054$, wild bootstrap $p = 0.053$. The full-sample coefficient falls short of conventional thresholds once small-cluster corrections are applied—a natural consequence of pooling events where the capability channel is active (closed-source) with those where it is attenuated (open-source).

The contrast with open-source releases supports H1: the appropriable capability channel operates specifically in settings where proprietary access is maintained.

4.2 Open-source attenuation

Table ?? examines whether open-source status moderates capability pricing by estimating:

$$\text{CAR}[0, +20]_{it} = \beta_1 \text{intel_}c_e + \beta_2 \text{open}_e + \beta_3 (\text{intel_}c_e \times \text{open}_e) + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_t + \varepsilon_{it} \quad (1)$$

where e indexes event, i firm, t year, and $\mathbf{X}_{it}$ contains firm-level controls.

The baseline (closed-source) slope $\hat{\beta}_1 = +0.0023$ (CR2 $p = 0.005$). The interaction coefficient $\hat{\beta}_3 = -0.0037$ (SE = 0.0012), implying a substantially attenuated open-source slope of -0.0015 . The interaction is significant under CR0 ($p = 0.003$) and marginally so under CR2 ($p = 0.048$); under the wild cluster bootstrap, however, $p = 0.086$, short of the conventional 5% threshold.

We interpret this evidence cautiously. The point estimates are consistent with H2—open-source status attenuates and in point estimates reverses the capability-pricing slope—but only 11 open-source events identify the interaction, and the wild bootstrap signals fragile inference at this cluster count. We treat this as suggestive evidence consistent with the appropriability mechanism, rather than a robust finding.

The open-source main effect at average intelligence ($\hat{\beta}_2 = -0.013$) is not statistically significant, consistent with the interaction story: open-source releases do not unconditionally yield lower returns; they yield lower *marginal returns to additional capability*.

4.3 Media sentiment and the hype alternative

Table ?? addresses whether the capability effect is confounded with media hype. Column (1) shows that sentiment alone strongly and *negatively* predicts $\text{CAR}[0, +20]$: $\hat{\beta} = -0.081$, CR2 $p = 0.003$. More positive media coverage in the event window is associated with *lower* subsequent 20-day returns. One interpretation is that events generating strong positive narratives have already experienced anticipatory price appreciation, so the $[0, +20]$ abnormal return is depressed by a prior run-up that our pre-event estimation window partially misses.

We cannot identify the precise mechanism with the current data.

Columns (2)–(3) show the joint model. Intelligence and sentiment both retain their sign and significance under CR0; under CR2, the intelligence coefficient is marginal ($p = 0.087$) and sentiment is marginal ($p = 0.057$). Crucially, controlling for sentiment does not eliminate the intelligence coefficient—the two variables are orthogonal at the event level (event-level regression of sentiment on intelligence: $p = 0.317$), confirming they capture distinct information. The interaction between intelligence and sentiment is not significant ($\hat{\beta} = +0.002$, $p = 0.555$), so sentiment neither amplifies nor dampens the capability channel.

5 Robustness

Alternative windows. Appendix Table A1 repeats the closed-source regression across windows from CAR[0, +1] to CAR[0, +20]. The coefficient is largest and most precisely estimated at the long window; the immediate [0, +1] response is positive but not significant, suggesting capability information takes time to be fully processed. This pattern is consistent with fundamental revaluation rather than noise trading around announcements.

Fama-French three-factor model. Replacing the market model with the Fama-French three-factor model leaves the main coefficients unchanged (Appendix Table A1).

Industry and Mag7 checks. Internet infrastructure, software, and semiconductor firms show the strongest positive response (CR2 $p \leq 0.015$ for the internet infrastructure subgroup). Results are not driven by the Magnificent Seven—the interaction between intelligence and a Mag7 indicator is not significant ($p = 0.633$), and the capability slope is nearly identical for Mag7 and non-Mag7 firms (Appendix Table A2).

Owner short-window effects. Firms that are direct owners of the releasing model show a negative short-window abnormal return (CAR[0, +1]: $\hat{\beta} = -0.0018$ for the *intel_c* $\times$ *owner*

interaction, wild $p = 0.034$). This pattern—in which the releasing firm’s equity does not benefit immediately—may reflect profit-taking or uncertainty about commercial adoption timelines. We treat this as suggestive and report full details in Appendix Table A3.

6 Conclusion

Financial markets price AI model capability selectively. Among closed-source releases, a one-standard-deviation increase in the Artificial Analysis Intelligence Index is associated with a 3.0-percentage-point higher 20-day cumulative abnormal return—a result that is robust to conservative small-cluster inference. Among open-source releases, the capability slope is attenuated and reverses sign in point estimates, though this interaction rests on only 11 open-source event clusters and should be treated as exploratory.

Media sentiment is negatively associated with post-release returns and operates independently of the capability channel, consistent with a mechanism in which positive narratives bring forward price gains that subsequently dissipate. Intelligence and sentiment are jointly significant in the full model, providing a rough decomposition of LLM event returns into a fundamental-capability component and a narrative component that move in opposite directions.

Financial markets appear to price the *economic ownership* of AI performance, not AI performance itself. As open-source LLMs mature and inference costs fall, the appropriability channel we identify may weaken, compressing the abnormal returns to AI capability improvements among affected firms.

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Tables

Table 1: Baseline Effect of AI Model Capability on Cumulative Abnormal Returns

	(1) Full sample	(2) Closed-source	(3) Full sample	(4) Closed-source
AA Intelligence Index (centered)	0.00152* (0.00067)	0.00232*** (0.00062)	0.00152	0.00232**
CR0 p -value	0.027	0.001		
CR2 p -value			0.054	0.007
Wild bootstrap p -value			0.053	0.008
Year FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Observations	3,780	2,899	3,780	2,899
Event clusters	47	36	47	36

Dependent variable is the market-model CAR[0, +20]. *intel_c* is the AA Intelligence Index centered at its sample mean (26.6). Columns (1)–(2) report CR0 clustered standard errors in parentheses; Columns (3)–(4) add CR2 and wild bootstrap p -values. Firm controls are log total assets, book-to-market, prior-quarter volatility, and 12-minus-1-month momentum. Wild bootstrap uses $B = 4,999$ Rademacher draws with the impose-null approach. $+p < 0.10$; $*p < 0.05$; $**p < 0.01$; $***p < 0.001$.

Table 2: The Role of Appropriability: Open-Source Interaction

	(1) CR0	(2) CR2
<i>intel_c</i> (closed-source slope)	0.00226*** (0.00060)	0.00226**
CR0 <i>p</i>	0.001	
CR2 <i>p</i>		0.005
<i>is_open_weight</i> (at mean intelligence)	−0.013 (0.012)	−0.013
<i>intel_c</i> × <i>is_open_weight</i>	−0.00373** (0.00120)	−0.00373*
CR0 <i>p</i>	0.003	
CR2 <i>p</i>		0.048
Wild bootstrap <i>p</i>	0.086	0.086
<i>Implied open-source slope</i> ($\hat{\beta}_1 + \hat{\beta}_3$)	−0.00148	−0.00148
Year FE	Yes	Yes
Firm controls	Yes	Yes
Observations	3,780	3,780
Event clusters	47	47

Dependent variable is CAR[0, +20]. Estimation equation is (??). CR0 standard errors in parentheses for Column (1). The interaction term is significant under CR0 and marginally so under CR2; the wild bootstrap *p*-value exceeds 0.05 and this result should be interpreted with caution. Other notes as in Table ??.

Table 3: Media Sentiment, Model Capability, and CAR[0, +20]

	(1) Sent. only	(2) Intel. only	(3) Joint	(4) Joint (CR2)
<i>sent_c5</i> (centered FinBERT)	−0.081*** (0.020)		−0.061* (0.026)	−0.061+
CR2 <i>p</i> (sent)	0.003			0.057
<i>intel_c</i>		0.00152* (0.00067)	0.00158* (0.00072)	0.00158+
CR2 <i>p</i> (intel)		0.054		0.087
Year FE	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Observations	3,068	3,780	3,068	3,068
Event clusters	47	47	47	47

sent_c5 is the FinBERT mean sentiment over a ± 5 -day event window, centered at its sample mean (0.18). Sentiment is available for approximately 3,068 observations. Column (4) reports CR2 *p*-values for both predictors; CR0 standard errors are in parentheses for Columns (1)–(3). The event-level correlation between *intel_c* and *sent_c5* is not statistically significant ($p = 0.317$), confirming orthogonality. Other notes as in Table ??.